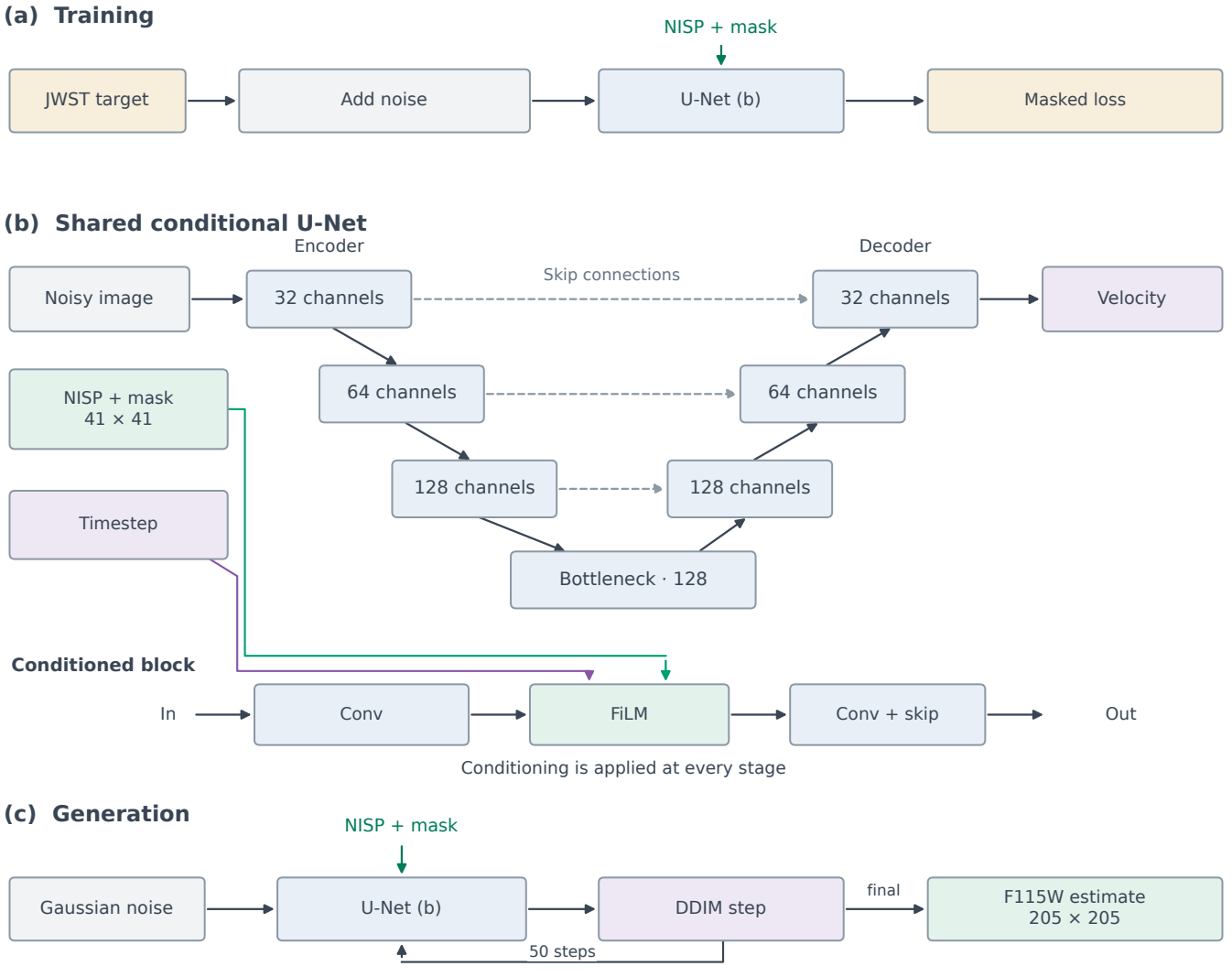
(a) Training
NISP + mask
JWST target
Add noise
U-Net (b)
Masked loss
(b) Shared conditional U-Net
Encoder
Decoder
Noisy image
32 channels
Skip connections
32 channels
Velocity
NISP + mask
41 × 41
64 channels
64 channels
128 channels
128 channels
Timestep
Bottleneck · 128
Conditioned block
In
Conv
FiLM
Conv + skip
Out
Conditioning is applied at every stage
(c) Generation
NISP + mask
Gaussian noise
U-Net (b)
DDIM step
final
F115W estimate
205 × 205
50 steps